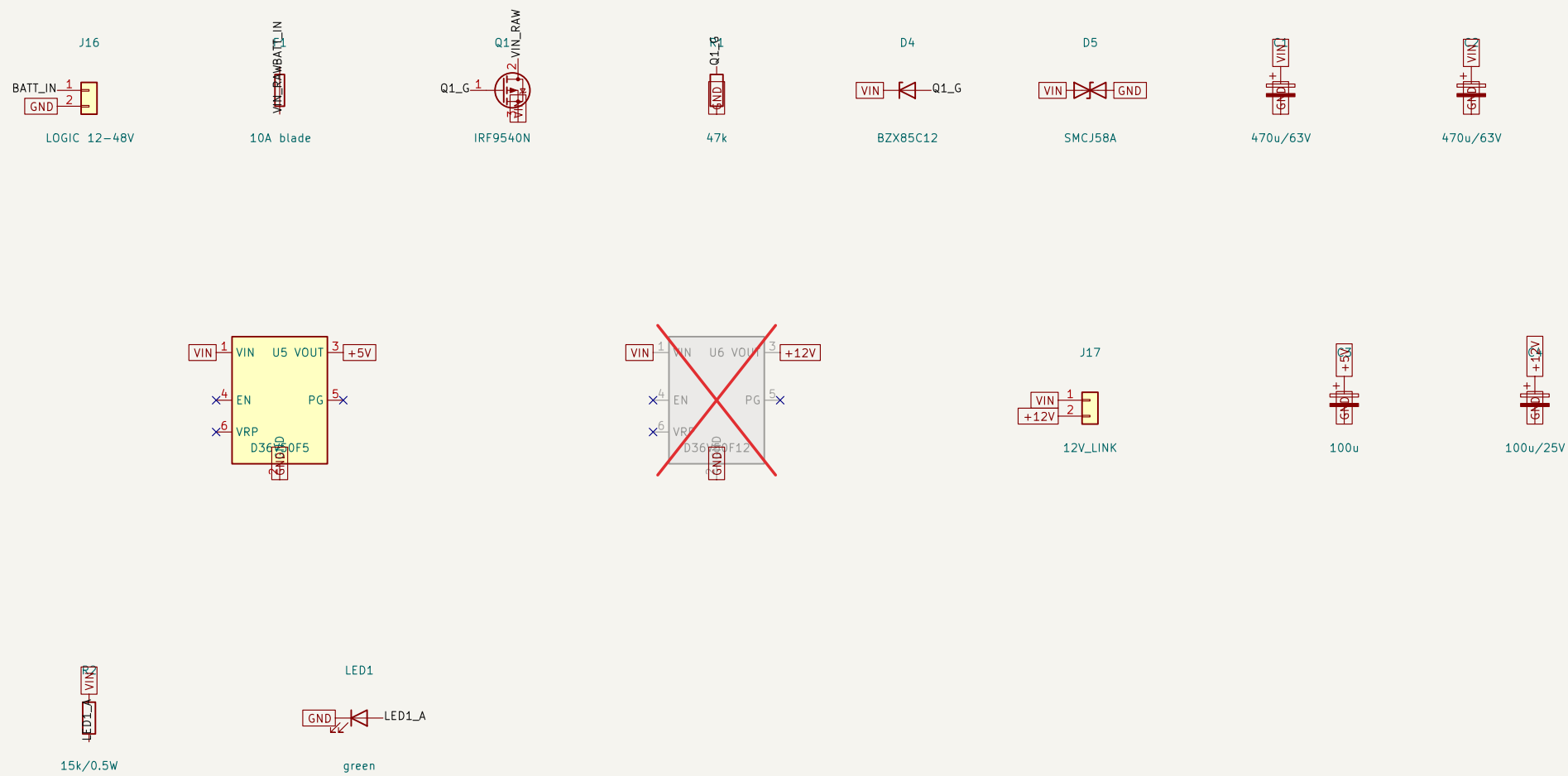


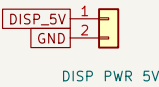
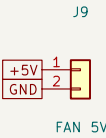
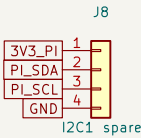
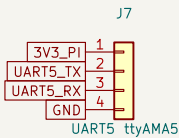
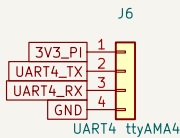
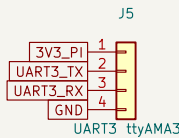
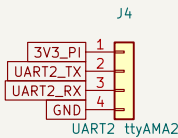
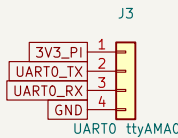
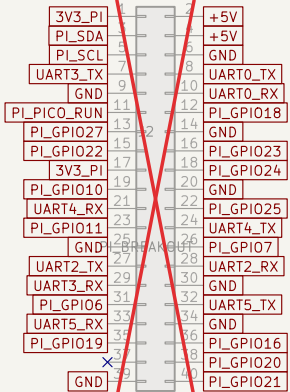
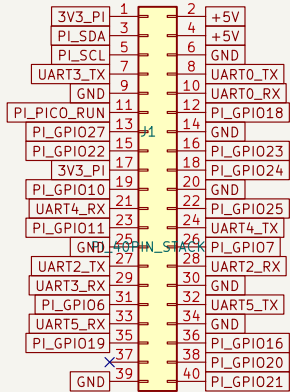


POWER: 12-48V battery in, protection, 5V/5A + 12V/4.5A rails  
12V-battery boats: omit U6 (DNP) and bridge J17 with a wire link

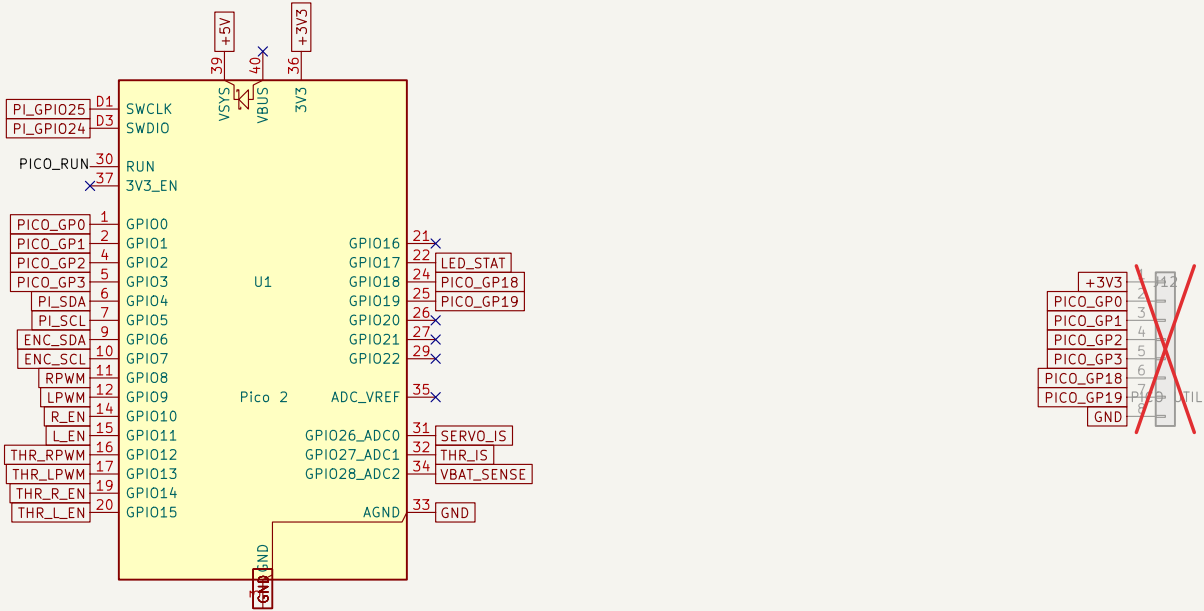


RASPBERRY PI 4/5 CARRIER: stacking header, UART breakouts, display power

DSI touchscreen: ribbon to Pi directly; 5V from J10. USB/HDMI on Pi itself.



MCU: Pico 2 (RP2350) – I2C0 slave @0x42 to Pi, I2C1 master to AS5600  
Firmware map: 0x00 CMD {pwm,dir,steer} / 0x10 STATUS {angle,ok,wrap,state,vbat,lsense}; 800ms watchdog  
Pico soldered directly (low profile, sits under the Pi). Pi reflashes it via SWD on GPIO24/25 (openocd).



1k



4.7k



4.7k



100k



6.8k



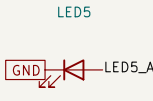
100n



BAT54S



1k



yellow



SERVO: on-board 2x BTN8982TA half-bridge (BTS7960-compatible interface), VM = +12V rail  
AS5600 (in servo housing per vanchor-ng cad/) cabled to J11; keep lead <1m, 100kHz  
24-48V boats: servo stall limited by BUCK2 (4.5A). 12V boats: J17 links VIN->12V, full stall current

